

SpectralOperator — network architecture

A coordinate-based operator that predicts log-flux density as a continuous function of energy, conditioned on plasma temperature. Two input streams stay separate — the **energy coordinate** builds the spectrum, the **temperature** shapes it — and meet only inside the trunk.

params 3.32 M

trunk 512 × 6

Fourier K 1024 · f≤4000

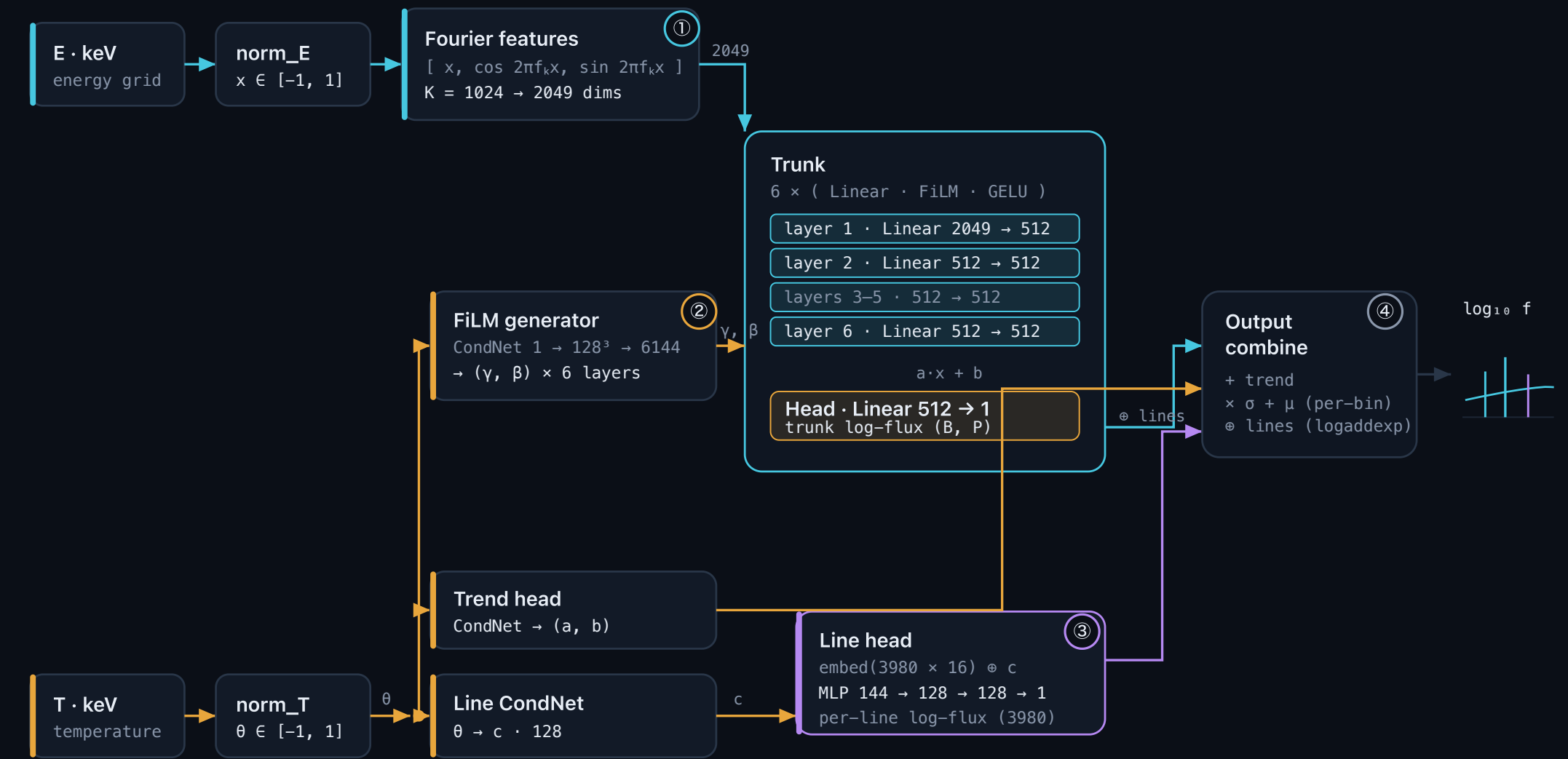
lines 3980

activation GELU

test MRE 0.13 %

t / spectrum 21.5 ms

energy / coordinate path temperature / conditioning path line structure ①–④ detail below



DETAIL ① · ENERGY ENCODING

Fourier feature embedding

A single scalar coordinate can't express one-bin emission lines. Each energy is lifted into 1024 sinusoids across four decades of frequency, so the trunk can resolve both the smooth continuum and sharp lines.



$x \mapsto [x, \cos 2\pi f_k x, \sin 2\pi f_k x]$
 f_k log-spaced, $f \in [0.25, 4000]$
curriculum $w_k = \text{clip}(\alpha K - k, 0, 1)$
→ high freqs fade in over first 30%

DETAIL ② · CONDITIONING

FiLM-modulated trunk layer

Temperature never enters as an input feature. Instead it generates a scale and shift for every trunk layer, steering the shared spectral backbone toward the spectrum at that T.



$h \leftarrow \text{GELU}(\gamma \odot W h + \beta)$
 $\gamma = 1 + \text{gen}(\theta)$ — identity at init
one (γ, β) pair per layer · 6 pairs total

DETAIL ③ · LINE STRUCTURE

Line-amplitude head

The 3980 known line energies are fixed. Each gets a learned embedding; an MLP turns [embedding, temperature code] into that line's flux, added to the continuum in *linear* space so lines never distort it.



$\text{out} = \text{logaddexp}(\text{trunk} \cdot \ln 10, \text{line} \cdot \ln 10) / \ln 10$
 $= \log_{10}(10^{\text{trunk}} + 10^{\text{line}})$ — add fluxes

DETAIL ④ · RECONSTRUCTION

Output combination

Four terms assemble the final prediction per energy bin. Per-bin normalisation carries the static line forest in its mean, so the trunk only has to learn the temperature-dependent deviations.



$f = (\text{head}(h) + a \cdot x + b) \cdot \sigma_{\text{bin}} + \mu_{\text{bin}}$
then for line bins: $\odot$ line head (logaddexp)

Shapes: **B** = batch of spectra, **P** = sampled energy points (all 24 212 native bins at inference; a random subset while training). The trunk is evaluated once per energy and shared across all temperatures — conditioning enters only through the amber path. Velocity broadening and instrument response are applied *after* the network, analytically, and are not part of it. Configuration shown is **big_trunk** (Tier-2 winner); the deployed **t04** family uses a 384 × 5 trunk and K = 512.